Homework #8

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ECON5345, Spring 2026
Due: April 12, 2026, 10:00 PM

Reminder: Empirical and simulation exercises must be submitted with "stand-alone" code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

1. Suppose that the central planner faces the following optimization problem:

$$\max E\sum_{t=0}^{\infty}\beta^t\left\{\frac{C_t^{1-\sigma}}{1-\sigma}-\frac{\xi L_t^{1+\eta^{-1}}}{1+\eta^{-1}}\right\}$$

subject to

$$Y_t = Z_t K_{t-1}^{\alpha} L_t^{1-\alpha}$$
$$Y_t = C_t + I_t + \frac{\phi}{2}\left(\frac{I_t}{K_{t-1}}-\delta\right)^2 K_{t-1}$$
$$K_t = (1-\delta)K_{t-1} + I_t$$
$$Z_t = e^{\epsilon_t} Z_{t-1}^{\rho}$$

Assume that $\rho < 1$ and $\epsilon_t \sim iid(0, \sigma_\epsilon^2)$. The structural parameters of the model are $(\sigma, \alpha, \xi, \eta, \delta, \rho, \phi, \sigma_\epsilon)$.

Part I: basic model

(a) Write the first order conditions (FOCs) for the model.
(b) Write the system of equations that needs to be solved to find the (non-stochastic) steady state values of endogenous variables.
(c) Log-linearize the model around the steady state.
(d) Which structural parameters can be estimated using the single-equation approach?

Part II: calibration and analysis of dynamic properties of the basic model

(e) Suppose that $\alpha = \frac{1}{3}$, $\eta = 1$, $\phi = 1$, $\delta = 0.025$, $\xi = 1$, $\beta = 0.99$, $\sigma = 1$, $\rho = 0.985$, $\sigma_\epsilon = 0.007$. Do these parameters roughly match steady states and great ratios we observe in the data? What data can be used to "estimate" ϕ?
(f) Using log-linearization from Part I, compute, plot and interpret impulse response for consumption, investment, output, capital, and labor to a unit shock in the level of technology (i.e., $\epsilon_t = 1$).
(g) Compute, plot and interpret impulse responses for alternative parameterizations: $\eta = 0.5, 10$; $\phi = 0,10$; $\sigma = 0.1,10$; $\rho = 0.5,0.9999$. Interpret the differences.
(h) Using the solution of the model, compute the variance of growth rates of the endogenous variables C_t, Y_t, L_t, and I_t as well as covariances of the growth rates of these variables and compare these statistics with the corresponding data analogues.
(i) Compare model impulse responses with impulse responses in identified VARs (e.g., Gali 1999).

Part III: Additional shocks and variance decomposition

(j) Now consider a modified model with government expenditures. Specifically, suppose that $Y_t = C_t + I_t + \frac{\phi}{2}\left(\frac{I_t}{K_{t-1}} - \delta\right)^2 K_{t-1} + G_t$ where $G_t = \kappa \exp(u_t)\, G_{t-1}^{\psi}$ and $u_t \sim iid(0, \sigma_u^2)$. Suppose that $\psi = 0.95$. Adjust κ to make the steady state G to Y ratio 0.2. Plot the impulse response of C_t, Y_t, L_t, K_t and I_t to a unit shock in G (i.e. $u_t = 1$)? Interpret your results. Contrast with the impulse responses to a technology shock.

(k) Suppose that $\sigma_u = 0.004$. Using your reduced-form solution $x_t = Ax_{t-1} + Bv_t$, compute and plot forecast variance decomposition (i.e., fractions due to technology and government spending shocks) for horizons h=0,1,…,40 for C_t, Y_t, L_t, and I_t.